# Chapter 14 | Emerging Trends (AI / ML / IoT / AR-VR)

## 14.1 Artificial Intelligence — Definition, Categories and Ethics

**Artificial Intelligence: making a machine intelligent** — getting it to perform tasks that would need intelligence if a person did them: reasoning, learning, perceiving, understanding language. It is not 'putting your intelligence into a computer' and certainly not adding more memory. Classic **applications: language understanding and problem solving (text analytics and NLP)**, expert systems, robotics, vision, game playing.

**(1) FOPL = First Order Predicate Logic** — the knowledge-representation logic extending propositional logic with objects, predicates and quantifiers (it can say 'all men are mortal'; propositional logic cannot).

**(2) Statistics for prediction**: **INFERENTIAL statistics helps make predictions from data** — it generalises from a sample to the population; descriptive statistics only summarises what was collected.

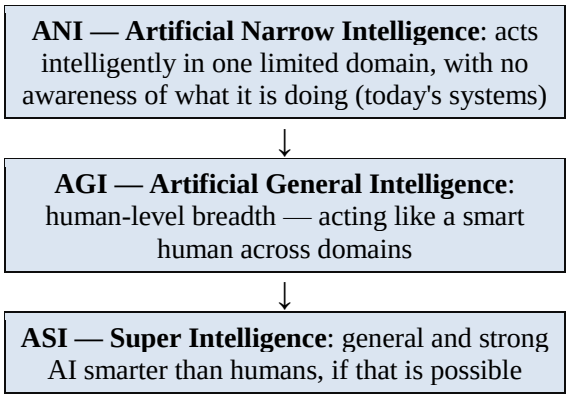



*Fig. 14.1 — The three categories of AI. The 2024 matching keyed I-2, II-3, III-1 on exactly these descriptions.*

**(3) AI ethics** (2024 matching, key I-2, II-1, III-3): **Data privacy = the trade-off between the amount of data collected and the performance** of a product or service; **K-anonymity = the standard metric measuring how many users in a database share a particular attribute**; **L-diversity extends that idea by merging several sensitive attributes**. And on **AI bias: many AI models are BLACK BOXES (only statement II)** — their internal decision-making is hard to interpret; deepfakes and K-anonymity are different concepts.

### Exam Points — 14.1

- AI = making a machine intelligent; application = language understanding/NLP; FOPL = First Order Predicate Logic; inferential statistics predicts from data.
- ANI narrow and unaware / AGI human-level / ASI beyond human (I-2, II-3, III-1); ethics matching: privacy = data-vs-performance trade-off, K-anonymity = shared-attribute metric, L-diversity merges sensitive attributes (I-2, II-1, III-3); AI-bias models = black boxes (only II).

## 14.2 Machine Learning and Deep Learning

| Type of learning | Data given | Idea |
|---|---|---|
| Supervised | Inputs WITH labels (the right answers) | Learns a mapping from examples — classification, regression |
| **Unsupervised** | **Inputs only — labels are NOT explicit** | **Looks for patterns or groupings in the data** (the asked definition) — clustering is the standard task |
| Reinforcement | Rewards and penalties from actions | Learns a policy by trial and error |

**(1)** **Deep learning — the vanishing gradient problem**: as the error is back-propagated through many layers, the **gradients become too small, slowing down learning** — the earliest layers barely change. (Exploding gradients are the too-large opposite; ReLU activations and residual connections are the usual remedies.)

**Exam Points — 14.2**

✦ Unsupervised learning = patterns/groupings without explicit labels; supervised has labels; reinforcement learns from rewards. Vanishing gradient = gradients too small → learning slows.

## 14.3 Expert Systems

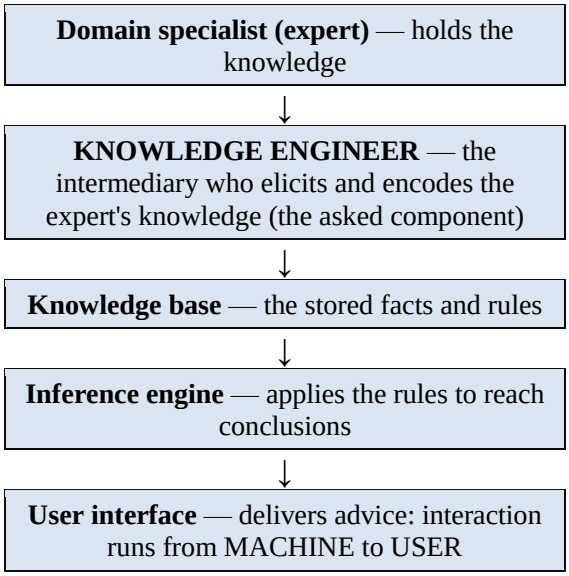



*Fig. 14.2 — Expert system architecture and the people around it.*

**(1)** **Characteristics** (2021 triple): the direction of interaction is **from machine to the user** (I true) and **its domain of knowledge is NARROW** (II true) — but 'information produced by analysis of operational and external data' describes an MIS/DSS, not an expert system (III false) → **I and II**.

**Exam Points — 14.3**

✦ Knowledge engineer = intermediary between specialist and system; characteristics = machine-to-user + narrow domain (I and II; operational-data analysis is MIS territory).

## 14.4 IoT, Industry 4.0 and AR/VR

**(1)** **IoT (Internet of Things)** connects everyday devices to the network. Its biggest adoption **challenge is SECURITY** — many weakly-protected devices form a large attack surface (deepfakes and black boxes are AI issues). The lightweight **device-communication protocol is MQTT** — publish/subscribe with tiny headers for low-power devices (HTTP and FTP are too heavy; CoAP and AMQP are the other IoT protocols). **IoT platforms: Amazon Web Services and Microsoft Azure — Flipkart is NOT one** (e-commerce).

**(2)** **Industry 4.0**: the era marked by **IoT, access to real-time data and cyber-physical systems** — the fourth industrial revolution (after steam, electricity/mass production, and computers).

**(3)** **AR/VR interfaces**: **HAPTIC technology provides TACTILE feedback** — forces, vibrations and motions that let the user feel virtual interactions rather than just see or hear them.

**Exam Points — 14.4**

✦ IoT challenge = security; IoT protocol = MQTT; platforms = AWS + Azure (Flipkart no); Industry 4.0 = IoT + real-time data + cyber-physical systems; haptic = tactile feedback.

### 14.5 Cloud Computing

**(1)** **Key features: scalability, self-service and pay-per-use — ALL THREE** (resources scale on demand, users provision them themselves, and pay only for use). **Examples of cloud computing: web applications, SaaS and Software-plus-Services — all options correct** (HTET 2025). The **major cloud platform in the UKPSC list is Microsoft Azure** (MS-Office is an application suite, Java a language; the big three: AWS, Azure, Google Cloud).

| Model | What you get | Asked fact |
|---|---|---|
| IaaS | Raw infrastructure — **a virtual machine with specified CPU, memory and disk** | **The bare VM is NOT PaaS** — the asked odd-one-out; it is Infrastructure as a Service |
| PaaS | Ready environment: OS + application software, web server, database server | The three genuine PaaS items in that question |
| SaaS | The finished application over the network | Software as a Service — a direct example of cloud computing |

#### Exam Points — 14.5

✦ Cloud features = scalability + self-service + pay-per-use (all); examples = web apps + SaaS + software-plus-services (all); Azure = major platform; a bare VM (CPU/memory/disk) = IaaS, NOT PaaS.

## Quick Revision — Chapter 14 (15 Points)

1. AI = making a machine intelligent (not adding memory); applications include language understanding, NLP, expert systems, vision.
2. FOPL = First Order Predicate Logic — objects, predicates, quantifiers.
3. Inferential statistics makes predictions from data; descriptive statistics only summarises.
4. AI ladder: ANI = narrow, acts without awareness; AGI = human-level general; ASI = smarter than humans (matching I-2, II-3, III-1).
5. AI ethics: data privacy = data-collected vs performance trade-off; K-anonymity = how many users share an attribute; L-diversity merges several sensitive attributes (I-2, II-1, III-3).
6. AI bias: many models are BLACK BOXES (only II) — opaque internal decisions.
7. Supervised = labelled data; UNSUPERVISED = patterns/groupings without explicit labels (clustering); reinforcement = rewards.
8. Vanishing gradient: gradients become too small → early layers stop learning; fixed by ReLU and residual connections.
9. Expert system: knowledge base + inference engine + user interface; the KNOWLEDGE ENGINEER is the intermediary between specialist and system.
10. Expert-system character: machine-to-user interaction, NARROW domain; analysing operational/external data is MIS, not an expert system.
11. IoT's main adoption challenge = SECURITY; MQTT is the lightweight IoT messaging protocol (publish/subscribe).
12. IoT platforms: AWS IoT, Azure IoT (Flipkart is e-commerce, not a platform).
13. Industry 4.0 = IoT + real-time data + cyber-physical systems (revolutions: steam → electricity → computers → cyber-physical).
14. Haptic technology = TACTILE feedback (forces and vibrations) for AR/VR.

**15.** Cloud: scalable, self-service, pay-per-use; SaaS/PaaS/IaaS — a bare virtual machine is IaaS; OS + web server + database server = PaaS; Azure among the big three platforms.

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